

The missing pressure traction in Scott–Vogelius–Nitsche methods on polygonal approximations of a curved wall

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Abstract

We consider two-dimensional Stokes flow in which a curved no-slip wall is replaced by an inscribed polygon with edges of length at most h_T . Following Gjerde and Scott, the flow is discretised on a mesh of size h_Ω by exactly divergence-free Scott–Vogelius elements of degree $k \geq 4$, with no-slip imposed on the polygon by Nitsche’s method with penalty μ/h_Ω . Scott asked whether the H^1 error is $O(h_T^{3/2} + h_\Omega^k)$ for general Stokes data, and if not, why not. For the method as printed and a fixed penalty, the answer is no whenever the pressure varies along the wall. The Nitsche form has no pressure traction, so the discrete flow leaks through the polygon with normal velocity $(h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean wall pressure. We prove that this leak is the leading error, with constant 1 in the slip and energy norms, and that the energy and H^1 errors are of order $(h_\Omega/\mu)^{1/2}$ and h_Ω/μ , with lower bounds. The H^1 rate is 1, not $1/2$, because incompressibility turns the normal part of the missing load into a tangential derivative. A mean-free traction term, whose velocity coincides with that of a zero-net-flux formulation of Frachon, Nilsson and Zahedi transcribed to this setting, attains $h_T^{3/2} + h_\Omega^k$ in H^1 for general data. A penalty scaling like $h_\Omega/\min_e |e|$ is sufficient, and necessary under a shape condition on one boundary triangle, by a witness certified in exact arithmetic. Computations with $k = 4$ agree with the predicted rates, constants and threshold.

Keywords: Stokes equations, Scott–Vogelius elements, Nitsche’s method, curved boundaries, divergence-free finite elements, penalty parameter

1. Introduction

For $k \geq 4$ on meshes without singular vertices, the divergence maps continuous piecewise- P_k velocities onto discontinuous piecewise- P_{k-1} pressures [1, 2], so Scott–Vogelius velocities are divergence-free pointwise. On fitted meshes with strong no-slip data this gives exact mass conservation and pressure-robust velocity errors [3]. At a curved wall the same property causes trouble. If the wall is replaced by an inscribed polygon and no-slip is imposed strongly, the constraints can force a zero velocity gradient at a polygon vertex (for instance one lying in only two triangles), giving $O(1)$ gradient errors there and an H^1 error of order $h^{1/2}$ [4, 5]; Section 4 shows that this depends on the mesh at the wall. Gjerde and Scott [5] keep the polygon and impose the wall condition by Nitsche’s method [6], with penalty μ/h_Ω , where h_Ω is the mesh size. Scott observed H^1 errors of order $h^{3/2}$ for a benchmark with zero pressure (since analysed in [7]). He asked for a proof of the error $h_T^{3/2} + h_\Omega^k$, where h_T is the longest polygon edge, for general Stokes data: “If this does not hold, why not?” [4].

The consistent Nitsche form for Stokes flow uses the full traction, so it contains $\langle p, v \cdot n \rangle$ (see, e.g., [8]), a term that is essential for consistency [9]. The form printed in [5, (27)–(28)] omits it. For exactly divergence-free spaces its placement matters. In both equations it spoils incompressibility near the boundary [10]. In the momentum equation alone [9, 11] it leaves the constant pressure mode free, and Frachon, Nilsson and Zahedi handle this by testing the momentum equation only with zero-net-flux velocities [11, §5.1]. Liu, Neilan and Otus [12] obtain an exactly

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divergence-free method inside a curved domain with a boundary multiplier, a zero-flux constraint and a Taylor correction. That method converges at order h^k , above the $h_\Gamma^{3/2}$ of the method below, which is sharp; our object is the Gjerde–Scott method.

Without a traction, only the penalty balances the wall pressure. The discrete flow therefore leaks through the polygon with normal velocity $(h_\Omega/\mu)(p - \bar{p})$, where $\bar{p}$ is the mean of p on the wall. This is the consistency error of penalty methods [13], acting on the pressure alone; as $h_\Omega/\mu \rightarrow 0$, the method still converges. *Contributions.* (i) We prove that the leak is the leading error, with constant 1 in the slip and energy norms, and give two-sided bounds of order $(h_\Omega/\mu)^{1/2}$ (energy) and h_Ω/μ (H^1) (Theorem 1). The H^1 rate is 1 because incompressibility turns the normal load into a tangential derivative (Lemma 3). So for fixed penalty, Scott’s estimate holds only when p is constant on the wall. (ii) We prove that the mean-free correction $\langle p_h - \bar{p}_{\Gamma_h}(p_h), v \cdot n \rangle$ attains $h_\Gamma^{3/2} + h_\Omega^k$ for general data and every $k \geq 4$ (Theorem 3). The device is not new: its velocity is that of the zero-net-flux formulation of [11], transcribed to this setting. The singularity of the naive form and the loss of incompressibility of the L_0^2 alternative (Proposition 2) are elementary; compare [11, (5.7)–(5.9)] and [12, Remark 3.2]. (iii) The penalty threshold scales like $\rho = h_\Omega / \min_e |e|$. This is sufficient, and it is necessary under a shape hypothesis, via a single-triangle witness certified in exact arithmetic (Proposition 4). (iv) A numerical study with $k = 4$ confirms the rates, the leak constant 1 and the penalty scaling; it includes the penalty study requested in the prize. We treat a disk in a square; Clough–Tocher refinements, general domains, Navier–Stokes, 3D and omitted proofs are in [14]. The analysis and computations were carried out with an AI research assistant, as declared at the end.

2. Setting and the three methods

Let $R = (-L, L)^2$, $L > 1$, D the unit disk, $\Omega = R \setminus \bar{D}$, $\Gamma = \partial D$, and let (u, p) be smooth with $-\Delta u + \nabla p = f$, $\operatorname{div} u = 0$ in Ω , $u|_\Gamma = 0$, $u|_{\partial R} = g$. P_h is a convex polygon with vertices on Γ , $\Gamma_h = \partial P_h$ and $\Omega_h = R \setminus P_h \supset \Omega$. On Γ_h and on Γ , n is the unit normal pointing out of Ω_h (resp. Ω); ν is the outward normal of R ; $\langle \cdot, \cdot \rangle$ is the $L^2(\Gamma_h)$ product. A triangulation of Ω_h with edges $\mathcal{E}_\Gamma$ on Γ_h has $h = h_\Omega := \max \operatorname{diam} T$, $h_\Gamma := \max_{\mathcal{E}_\Gamma} |e|$, $\rho := h / \min_{\mathcal{E}_\Gamma} |e|$ and $\gamma := \mu h_\Gamma / h$ (the effective boundary penalty, $\mu/h = \gamma/h_\Gamma$). We assume (M0) shape regularity and $|e| \geq c_0 h_\Gamma$ on $\mathcal{E}_\Gamma$, (M1) no singular vertices, and (M2) the angle condition of [2, Thm. 1]; $h_\Gamma \ll h$ is allowed. For $k \geq 4$, V_h is the space of continuous piecewise- P_k fields, V_h^R and V_h^0 are its subspaces vanishing on ∂R and on $\partial\Omega_h$, Π_h is the space of discontinuous piecewise- P_{k-1} functions, and $Z_h := \{v \in V_h^R : \operatorname{div} v = 0\}$. Then $\operatorname{div} V_h^R = \Pi_h$, and $(V_h^0, \Pi_h \cap L_0^2)$ has an inf-sup constant β independent of h [2], [14, Lemmas 4.3, 4.5]. With $Du = \nabla u + (\nabla u)^\top$, $\partial_n u = (\nabla u)n$ and $\bar{p}_{\Gamma_h}(q) := |\Gamma_h|^{-1} \int_{\Gamma_h} q$, the Nitsche form

$$N_h(u, v) = \frac{1}{2}(Du, Dv) - \langle \partial_n u, v \rangle - \langle u, \partial_n v \rangle + (\mu/h)\langle u, v \rangle$$

is the symmetric Nitsche form of [5]. All methods seek $u_h \in V_h$ with the outer data on ∂R and $p_h \in \Pi_h$ with $(q, \operatorname{div} u_h) = 0$ for $q \in \Pi_h$ and

$$N_h(u_h, v) - (p_h, \operatorname{div} v) + b_\Gamma(p_h, v) = (\tilde{f}, v) \quad \forall v \in V_h^R. \quad (1)$$

The printed method GS [5, (28)] has $b_\Gamma = 0$ (we read its test space as V_h^R and use the opposite pressure sign; see [14, §2]). The naive form has $b_\Gamma = \langle p_h, v \cdot n \rangle$, the consistent traction term. The mean-free form CNS* has $b_\Gamma = \langle p_h - \bar{p}_{\Gamma_h}(p_h), v \cdot n \rangle$. The L_0^2 variant is the naive form with $p_h, q \in \Pi_h \cap L_0^2(\Omega_h)$. The outer data are the P_k interpolant g_I of g , or $\tilde{g}_I := g_I - (m_R/8L^2)x$ with $m_R := \int_{\partial R} g_I \cdot \nu$, which has zero net flux, like the exact data (cf. [15]). The velocities of GS and CNS* are divergence-free pointwise.

To compare u_h with u on $\Omega_h \supset \Omega$ we extend u as a divergence-free field. Let $\tilde{u} := \operatorname{curl} \tilde{\psi}$ for a smooth extension $\tilde{\psi}$ of the stream function of u , let $\tilde{p}$ and $\tilde{f}$ extend p and f , and let $F := \tilde{f} + \Delta \tilde{u} - \nabla \tilde{p}$, which vanishes on Ω . Set $\|v\|^2 := \|\nabla v\|^2 + \frac{\mu}{h}\|v\|_{\Gamma_h}^2 + \sum_{e \in \mathcal{E}_\Gamma} |e| \|\partial_n v\|_{L^2(e)}^2$ and $G := \|p - \bar{p}\|_{L^2(\Gamma)}$. Let $\varepsilon_h := h^k + \inf [\tilde{u} - z]$, the infimum over divergence-free $z \in V_h$ with the outer data. For bounded γ , $\varepsilon_h \leq Ch^k$ with data $\tilde{g}_I$, and also with g_I if $h_\Gamma \geq h^{2(\sigma_k - k)}$, where $\sigma_k = k + 2$ (k even) or $k + 1$ (k odd) [14, Lemmas 4.6, 7.21]. Constants C, c do not depend on $h, h_\Gamma, \mu, \gamma, \rho$; C_p may also depend on $\|p - \bar{p}\|_{W^{1,\infty}(\Gamma)}$.

Lemma 1 ([14, Lemma 4.2]). *Let κ be the Korn constant on V_h^R (uniform in h) and C_I the inverse trace constant. If $\mu \geq 4\kappa C_I^2 \rho$, which holds when $\gamma \geq \gamma_0 := 4\kappa C_I^2 / c_0$, then $N_h(v, v) \geq c_1 \|v\|^2$ and $|N_h(w, v)| \leq M \|w\| \|v\|$ on V_h^R , with c_1, M independent of h, μ, ρ .*

3. Main results

Lemma 2 (Consistency). *For $v \in V_h^R$, $N_h(\tilde{u}, v) - (\tilde{p}, \operatorname{div} v) + \langle \tilde{p}, v \cdot n \rangle = (\tilde{f}, v) + \mathcal{G}(v)$, where $|\mathcal{G}(v)| \leq C(1 + \gamma^{1/2})h_\Gamma^{3/2}|v|$ and $\mathcal{G}(v) := \langle (\nabla \tilde{u})^\top n, v \rangle - \langle \tilde{u}, \partial_n v \rangle + \frac{\mu}{h} \langle \tilde{u}, v \rangle - (F, v)$.*

Proof. Green's formula gives the identity. Chords lie within $h_\Gamma^2/4$ of Γ , so $\tilde{u} = O(h_\Gamma^2)$ on Γ_h and $|(F, v)| \leq Ch_\Gamma^2 \|\nabla v\|$, which bounds the last three terms. On Γ , $(\nabla u)^\top n = n \operatorname{div} u = 0$. On a chord e the normal tilts by an angle $s + O(h_\Gamma^2)$, where s is the arclength from the midpoint m_e , so $(\nabla \tilde{u})^\top n = s a_e + O(h_\Gamma^2)$ with a_e constant on e . The remainder contributes $Ch_\Gamma^2 \|v\|_{L^1(\Gamma_h)}$. The leading term is $O(h_\Gamma)$ pointwise but odd in s : $|\int_e s a_e \cdot v| = |\int_e s a_e \cdot (v - v(m_e))| \leq |e|^2 \|\partial_\tau v\|_{L^1(e)}$, and an inverse trace inequality gives $Ch_\Gamma^{3/2} \|\nabla v\|$. $\square$

For $v \in Z_h$, $\int_{\Gamma_h} v \cdot n = \int_{\Omega_h} \operatorname{div} v = 0$, so for GS

$$N_h(\tilde{u} - u_h, v) = \mathcal{R}(v) + \mathcal{G}(v), \quad \mathcal{R}(v) := -\langle \tilde{p} - \bar{p}, v \cdot n \rangle \quad \forall v \in Z_h. \quad (2)$$

To identify the response to the missing load $\mathcal{R}$, let $\varphi := \chi(r) \int_0^\theta (p - \bar{p}) d\theta'$ in polar coordinates, with a cut-off χ equal to 1 near $r = 1$ and to 0 near ∂R ; then $w := \operatorname{curl} \varphi = -(p - \bar{p})n$ on Γ . With $I_h^{C^1}$ the C^1 piecewise- P_{k+1} interpolant of [16], $v_\varphi := \operatorname{curl} I_h^{C^1} \varphi \in Z_h$, $\|v_\varphi - w\|_{W^{1,\infty}} \leq Ch^k$, and $v_\varphi = -(p - \bar{p})(x^*)n + O(h_\Gamma)$ on Γ_h , where x^* is the radial projection of x onto Γ .

Theorem 1 (The printed method [14, §6]). *Let $\gamma \geq \gamma_0$ and $Y := (1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h$. (a) $\|\tilde{u} - u_h\| \leq C[(h/\mu)^{1/2}G + Y]$. If $G > 0$ and $h \leq h_0(G)$, with h_0 independent of γ , then $\|\tilde{u} - u_h\| \geq (2M)^{-1}(h/\mu)^{1/2}G - CY$. (b) $\|\nabla(\tilde{u} - u_h)\| \leq C_p h/\mu + CY$. (c) Let $G > 0$, fix γ , and let $h \rightarrow 0$ with ρ bounded and $\varepsilon_h \leq Ch^{3/2}$. Then $\tilde{u} - u_h = (h/\mu)v_\varphi + \vartheta_h$ with $\|\vartheta_h\| = o((h/\mu)^{1/2})$ and $\|\vartheta_h\|_{\Gamma_h} = o(h/\mu)$; hence*

$$\frac{\|\tilde{u} - u_h\|_{L^2(\Gamma_h)}}{(h/\mu)G} \rightarrow 1, \quad \frac{\|\tilde{u} - u_h\|}{(h/\mu)^{1/2}G} \rightarrow 1, \quad c \frac{h}{\mu} G \leq \|\nabla(\tilde{u} - u_h)\| \leq C_p \frac{h}{\mu}.$$

The key to (b) is the following lemma; an inverse estimate would only give $|\langle S, \partial_n v \rangle| \leq Ch_\Gamma^{-1/2} \|\nabla v\|$.

Lemma 3 (Normal-load cancellation). *Let τ_e and n_e be the unit tangent and normal of $e \in \mathcal{E}_\Gamma$. Let S be edgewise $W^{1,\infty}$ on Γ_h and continuous at its vertices, with $|S \cdot \tau_e| \leq Ch_\Gamma$ on each e . Then $|\langle S, \partial_n v \rangle| \leq C(\|v\|_{\Gamma_h} + h_\Gamma^{1/2} \|\nabla v\|) \leq C \|\nabla v\|$ for $v \in Z_h$.*

Proof. The tangential part is at most $Ch_\Gamma \|\partial_n v\|_{\Gamma_h} \leq Ch_\Gamma^{1/2} \|\nabla v\|$. On e , $\operatorname{div} v = 0$ gives $\partial_n v \cdot n_e = -\partial_\tau(v \cdot \tau_e)$, so with $\sigma_e := S \cdot n_e$, $\int_e \sigma_e \partial_n v \cdot n_e = \int_e \partial_\tau \sigma_e (v \cdot \tau_e) - [\sigma_e v \cdot \tau_e]_{\partial e}$. The integral is at most $C\|v\|_{L^1(e)}$. At a vertex x shared by e and e' the endpoint terms combine to $v(x) \cdot (\sigma_{e'} \tau_{e'} - \sigma_e \tau_e) = O(|e| + |e'|)|v(x)|$, since S is continuous and the edge directions differ by $O(|e| + |e'|)$. The one-dimensional inverse inequality $\|v\|_{L^\infty(e)} \leq C|e|^{-1/2} \|v\|_{L^2(e)}$ for polynomials gives $\sum_e |e| \|v\|_{L^\infty(e)} \leq C|\Gamma_h|^{1/2} \|v\|_{\Gamma_h}$, and $\|v\|_{\Gamma_h} \leq C \|\nabla v\|$ uniformly in h [14, Lemma 3.2]. $\square$

Proof of Theorem 1. Since the projection $\Gamma_h \rightarrow \Gamma$ has Jacobian $1 + O(h_\Gamma^2)$, $|\mathcal{R}(v)| \leq (G + Ch_\Gamma^2)(h/\mu)^{1/2} |v|$. (a) Test (2) with $z - u_h \in Z_h$, z attaining ε_h , for the upper bound. For the lower, $\mathcal{R}(v_\varphi) = G^2 + O(h_\Gamma^2 + h^k)$, $|v_\varphi| \leq (\mu/h)^{1/2}(G + Ch_\Gamma^2 + Ch^k) + C$, and $M\|\tilde{u} - u_h\| \geq (\mathcal{R}(v_\varphi) - |\mathcal{G}(v_\varphi)|)/|v_\varphi|$. (b) Write $z - u_h = \delta_R + \delta_G$ in Z_h with $N_h(\delta_R, \cdot) = \mathcal{R}$ and $N_h(\delta_G, \cdot) = \mathcal{G} - N_h(\tilde{u} - z, \cdot)$ on Z_h , so that $|\delta_G| \leq CY$. For $r_h := \delta_R - (h/\mu)v_\varphi$ and $v \in Z_h$,

$$N_h(r_h, v) = -\langle (\tilde{p} - \bar{p})n + v_\varphi, v \rangle - \frac{h}{\mu} \left[\frac{1}{2} (Dv_\varphi, Dv) - \langle \partial_n v_\varphi, v \rangle - \langle v_\varphi, \partial_n v \rangle \right].$$

On e , $(\tilde{p} - \bar{p})n + v_\varphi = (p - \bar{p})(m_e^*) \tau_e + O(h_\Gamma^2 + h^k)$, with m_e^* the projection of m_e . This is odd in s to leading order, so the first term is at most $C_p(h_\Gamma^{3/2} + h^k) \|\nabla v\|$, as in Lemma 2. Lemma 3 with $S = w$ gives $|\langle v_\varphi, \partial_n v \rangle| \leq (C_p + Ch^k h_\Gamma^{-1/2}) \|\nabla v\|$. Testing with $v = r_h$ gives $|r_h| \leq C_p(h/\mu + h_\Gamma^{3/2}) + C\varepsilon_h$, and (b) follows. (c) Here $\vartheta_h = (\tilde{u} - z) + \delta_G + r_h$, so $\|\vartheta_h\| \leq C_p h/\mu + CY$, $\|\vartheta_h\|_{\Gamma_h} \leq (h/\mu)^{1/2} \|\vartheta_h\|$, and $\|v_\varphi\|_{\Gamma_h} = G + O(h_\Gamma^2 + h^k)$. Testing (2) with v_φ , all terms other than $(\mu/h) \langle \tilde{u} - u_h, v_\varphi \rangle$ and $\mathcal{R}(v_\varphi)$ are $o(G^2)$ by (a), (b), so $\|\tilde{u} - u_h\|_{\Gamma_h} \geq c(h/\mu)G$, and a uniform trace inequality gives $\|\tilde{u} - u_h\|_{\Gamma_h} \leq C \|\nabla(\tilde{u} - u_h)\| + Ch^{k+1/2}$. $\square$

Proposition 2 (Naive and L_0^2 forms). (i) The naive system is square, and $(0, 1)$ lies in the kernel of its matrix. (ii) Every solution of the L_0^2 variant has $\operatorname{div} u_h \equiv c$ with $c|\Omega_h| = \int_{\partial R} g_I \cdot \nu + \int_{\Gamma_h} u_h \cdot n$. It solves the naive system if and only if $c = 0$, and then, for $\gamma \geq \gamma_2$, u_h is the CNS* velocity.

Proof. (i) $-(1, \operatorname{div} v) + \langle 1, \nu \cdot n \rangle = 0$ for $v \in V_h^R$. (ii) $\operatorname{div} u_h \in \Pi_h$ is orthogonal to $\Pi_h \cap L_0^2$, hence constant. For $q \in L_0^2$ the naive pressure form equals the CNS* form at $q - \bar{p}_{\Gamma_h}(q)$, by (i). If $c = 0$, then $(u_h, p_h - \bar{p}_{\Gamma_h}(p_h))$ solves CNS*, whose solution is unique by Theorem 3. $\square$

Under weak imposition nothing forces $c = 0$, even for $\tilde{g}_I$. In our runs the L_0^2 variant was uniquely solvable with $c \neq 0$ (test A below, $N = 16$: $\|\operatorname{div} u_h\|_{L^2(\Omega_h)} = 4.5 \cdot 10^{-3}$, against $4 \cdot 10^{-13}$ for CNS*), so the naive system had no solution.

Theorem 3 (The mean-free method; [14, Thm. 7.2, Rem. 7.3]). There is $\gamma_2 \asymp \max(\gamma_0, \beta^{-2})$ such that for $\gamma \geq \gamma_2$: (a) CNS* is uniquely solvable, and its velocity equals that of the naive form tested only with $\{v \in V_h^R : \int_{\Gamma_h} v \cdot n = 0\}$, as in [11]; (b) $\|\tilde{u} - u_h\| \leq C[(1 + \gamma^{1/2})h_\Gamma^{3/2} + \varepsilon_h]$; (c) consequently, for $\gamma \leq \gamma_{\max}$ and data $\tilde{g}_I$, $\|\tilde{u} - u_h\|_{H^1(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$, with C depending on $\gamma_{\max}$ (the same holds with g_I if $h_\Gamma \geq h_\Omega^{2(\sigma_k - k)}$).

Sketch. Let $B^*(q, v) := -(q, \operatorname{div} v) + \langle q - \bar{p}_{\Gamma_h}(q), \nu \cdot n \rangle$. Then $B^*(q + c, v) = B^*(q, v) - c \int_{\Gamma_h} \nu \cdot n$, and Lemma 2 gives $N_h(e_u, v) + B^*(e_p, v) = \mathcal{G}(v)$ on V_h^R , where $e_u := \tilde{u} - u_h$, $e_p := p^* - p_h$ and $p^* := \tilde{p} - \bar{p}_{\Gamma_h}(\tilde{p})$. Write $e_p = \eta + \xi$ with $\eta := p^* - \pi_h p^*$, π_h the L^2 projection onto Π_h , and let $\bar{\xi}$ be the mean of ξ over Ω_h . Testing with V_h^0 and using the inf-sup condition gives $\beta\|\xi - \bar{\xi}\| \leq C(\|e_u\| + h_\Gamma^{3/2})$. Testing with $z - u_h \in Z_h$ leaves $\langle \eta + \xi - \bar{\xi}, (z - u_h) \cdot n \rangle$, and $|\langle \xi - \bar{\xi}, \nu \cdot n \rangle| \leq C(c_0 \gamma)^{-1/2} \|\xi - \bar{\xi}\| \|v\|$ by a discrete trace inequality; this is absorbed for $\gamma \geq \gamma_2$, which proves (b). With zero data the same argument gives $u_h = 0$ and $p_h \equiv c$, and an edge bubble with nonzero flux gives $c = 0$. On zero-flux test functions B^* is the naive form, and adding a constant turns a zero-flux solution into a CNS* solution; this proves (a). (c) follows with a Friedrichs inequality. $\square$

Remark 1 (Further results, proved in [14]). (1) *Pressure of GS* [14, Def. 8.10, Thm. 8.11]. Let $G > 0$, fix γ , let ρ be bounded and $h \leq h_0(G)$. Then $ch_\Gamma^{1/2}G/\gamma - C_p h_\Gamma \leq \min_{c' \in \mathbb{R}} \|p^* - p_h - c'\| \leq \|p^* - p_h\| \leq Ch_\Gamma^{1/2}G/\gamma + C_p h_\Gamma$. The error sits in the triangles along Γ_h , where it does not converge pointwise. (2) *CNS* uniformly in γ , and its pressure* [14, Thms. 7.19, 7.22, 8.7, Lemma 8.3]. With data $\tilde{g}_I$, Theorem 3(c) holds with C independent of $\gamma \geq \gamma_2$, and, for $\gamma \leq \gamma_{\max}$, $\|p^* - p_h\|_{L^2(\Omega_h)} \leq C(h_\Gamma^{3/2} + h_\Omega^k)$ with C depending on $\gamma_{\max}$. (3) *Sharpness* [14, Thm. 7.13]. If $W = \partial_n u \cdot \tau \neq 0$ on Γ , then $\|\nabla(\tilde{u} - u_h)\| \geq c\|W\|h_\Gamma^{3/2} - Ch_\Gamma^2$ for CNS*, and for GS with any G , c independent of γ, ρ, p , for $k \geq 7$; for $4 \leq k \leq 6$ under a local condition on boundary stars, verified numerically on our meshes but not proved. (4) *A growing penalty* [14, Thm. 7.19, Cor. 7.23, Thm. 7.24, Prop. 7.26]. For GS with data $\tilde{g}_I$, $\|\nabla(\tilde{u} - u_h)\| \leq C_p h_\Gamma/\gamma + C(h_\Gamma^{3/2} + h_\Omega^k)$ uniformly in $\gamma \geq \gamma_0$, so $\mu \gtrsim hh_\Gamma^{-3/2}$ restores the H^1 rate $3/2$. The energy norm cannot be rescued: if $W \neq 0$, $\gamma \geq \gamma_1$, $h_\Gamma \leq h_1$ and $h^k \leq h_\Gamma$, then $\|\tilde{u} - u_h\| \geq c\|W\|\gamma^{1/2}h_\Gamma^{3/2} - C(h_\Gamma/\gamma)^{1/2}$, and with Theorem 1(a) no penalty scaling gives an energy rate above 1. On quasi-uniform meshes the velocity block and pressure Schur complement condition numbers grow by $1 + \gamma$.

Proposition 4 (Penalty threshold [14, §9]). (a) $\mu \geq 4\kappa C_I^2 \rho$ suffices (Lemma 1). (b) Let T have an edge $e \subset \Gamma_h$ and not meet ∂R . Map it by a similarity to the triangle with base $[(0, 0), (1, 0)]$ and apex (a, b) , with barycentric coordinates λ_0 (apex) and λ_1, λ_2 . For $t \in \mathbb{R}^3$ let $v_t := \operatorname{curl}(\lambda_1^2 \lambda_2^2 (t_0 \lambda_0 + t_1 \lambda_1 + t_2 \lambda_2)) \in P_4^2$. Let A_T, B_T, C_T be the quadratic forms $\int_T \frac{1}{2} |Dv_t|^2$, $\int_e \partial_n v_t \cdot \nu_t$ and $\int_e |v_t|^2$, and let λ_T be the top eigenvalue of the pencil $(2B_T - A_T, C_T)$. If $|\mu|e/h < \lambda_T$, then N_h is indefinite on Z_h for every $k \geq 4$. (c) $\lambda_T > 1$ if the apex angle is at least 35° and the base angles at least 5° ; $\lambda_T > 4$ and $\lambda_T > 9.3$ for apex angles at least 45° and 60° . So if the triangle on a shortest edge of Γ_h has such a shape, coercivity on Z_h requires $\mu > \rho$.

Proof. (b) The potential vanishes to second order on the edges through the apex, so v_t , extended by zero, lies in Z_h . A_T and B_T are invariant under dilation and C_T scales with $|e|$, so $N_h(v_t, v_t) = (\mu|e|/h - \lambda_T)C_T(t) < 0$ for the top eigenvector. (c) A_T, B_T, C_T are explicit rational functions of (a, b) [14, §9.1]. On dyadic rational boxes covering each shape region we verify $2520b^3q^\top(2B_T - A_T - \lambda C_T)q > 0$ for a rational q in exact interval arithmetic (297 boxes for $\lambda = 1$) [14, Prop. 9.3]. The same test fails for $\lambda = 9.4$ at 60° , as it must. For tall triangles $\lambda_T < 0$; necessity under (M0)–(M2) alone is open [14, Rem. 9.5]. $\square$

Table 1: H^1 seminorm errors (rate), tests A and B₁₀₀, and pressure errors $\|p^* - p_h\|_{L^2(\Omega_h)}$ for B₁₀₀. Strong: $u_h = 0$ at the nodes of Γ_h , on mesh (a) (alternating first-layer diagonals; polygon vertices in two or four triangles, violating (M2)). GS, CNS*: standard mesh, three triangles per polygon vertex. Rates are from the preceding level of $N = 16, 24, 32, 48, 64, 96, 128, 192, 256$.

N	test A, velocity			test B ₁₀₀ , velocity		test B ₁₀₀ , pressure	
	strong (a)	GS	CNS*	GS	CNS*	GS	CNS*
32	5.72e-1 (0.72)	5.55e-2 (1.79)	6.41e-2 (1.80)	3.51 (0.93)	2.34e-2 (2.43)	18.2 (0.68)	7.67e-2 (3.03)
64	3.75e-1 (0.62)	1.81e-2 (1.68)	2.08e-2 (1.69)	1.88 (0.97)	6.83e-3 (1.76)	11.8 (0.65)	1.62e-2 (2.08)
96	2.98e-1 (0.59)	9.57e-3 (1.63)	1.09e-2 (1.64)	1.28 (0.98)	3.54e-3 (1.68)	9.23 (0.62)	8.01e-3 (1.80)
128	2.54e-1 (0.56)	6.11e-3 (1.59)	6.96e-3 (1.60)	9.71e-1 (0.98)	2.24e-3 (1.63)	—	—
192	—	3.27e-3 (1.57)	3.71e-3 (1.58)	6.54e-1 (0.99)	1.18e-3 (1.60)	—	—
256	—	2.10e-3 (1.55)	2.38e-3 (1.56)	—	—	—	—

Table 2: Coercivity threshold μ^* and $\gamma^* = \mu^* h_\Gamma / h$, with h/h_Γ and ρ varied through the outer box $(-L, L)^2$ and Γ_h fixed. Under refinement at $L = 2.5$, $\gamma^* = 14.7, 18.1, 19.9, 20.8$ for $N = 8, 16, 32, 64$.

N = 16	L	2.5	5	10	20	40	N = 32	2.5	5	10	20	40
	ρ	4.70	8.47	16.2	31.8	63.0		5.73	9.98	18.7	36.3	71.5
	μ^*	59.3	112	208	411	809		66.0	114	218	422	836
	γ^*	18.1	19.0	18.4	18.5	18.4		19.9	19.7	20.1	20.0	20.1

4. Numerical study

We take $k = 4$, $L = 2.5$, and N polygon vertices at the radial projections of N equispaced points on the square; then $h/h_\Gamma \approx 3.3$, $\max |e|/\min |e| \leq 1.97$, and each vertex of Γ_h lies in three triangles. Direct solves give relative residuals below 10^{-12} and $\|\text{div } u_h\| \leq 2 \cdot 10^{-9}$ (10^{-6} for the nearly singular strong solves on mesh (a)). We use $\mu = 100$ ($\gamma \approx 30$), known to be above the measured coercivity threshold only. Test A is Scott's benchmark $u = (1 - r^{-2})(-y, x)$, $p = 0$. Test B_a has stream function $(r^2 - 1)^2(x + y^2/2)/10$ and $p = a(x^2y - y + x/2)$, so $G = a(7\pi/8)^{1/2}$.

In Table 1 strong imposition locks (rate $\rightarrow 1/2$), while on test A ($G = 0$) both Nitsche methods tend to $3/2$. On our standard mesh strong imposition does not lock (rate 1.61 at $N = 128$). On B₁₀₀ the printed method has H^1 rate 1, an error over 500 times that of CNS* at $N = 192$, pressure rate decreasing towards $1/2$, and a maximum pressure error that does not decrease (77 at $N = 96$). The CNS* rates decrease monotonically towards $3/2$; its errors for B₁ and B₁₀₀ agree to all printed digits, and with the non-polynomial pressure $p = e^{x/2} \cos y$ its H^1 error differs from that for B₁ by at most $8 \cdot 10^{-7}$ relative. On the pure-pressure test ($u = 0$, $f = \nabla p$, p as in B₁), CNS* gives $u_h = 0$ to round-off, and GS reproduces the predicted leak. At $N = 96$ the normalised slip and energy errors of Theorem 1(c) are 0.981 and 0.996 for $\mu = 10^2$; for $\mu = 10^4$, which lies outside the regime of Theorem 1(c), they are 0.9993 and 0.9994.

We locate μ^* by bisection on the inertia of the velocity block augmented by a divergence penalty. With Γ_h fixed and the far field coarsened (Table 2), μ^* is linear over more than a decade, with $\gamma^* \approx 18$ –21 and $\mu^*/\rho \approx 11$ –13. This sweep keeps $\max |e|/\min |e|$ fixed, so it cannot separate ρ from h/h_Γ ; under refinement μ^*/ρ drifts from 14.7 to 11.1 while γ^* drifts from 14.7 to 20.8. The certified $\lambda_T > 9.3$ is a local lower bound consistent with these values; the gap comes from the flatter first layer and from collective boundary modes [14, §9.1]. On these meshes ($k = 4$), $\mu \geq 25\rho$, which implies $\gamma \geq 25 > \gamma^*$, was always coercive, while a fixed μ fails once $h/h_\Gamma \gtrsim \mu/20$.

Declaration of competing interest

The question studied is the subject of a prize offered by L. R. Scott; the author intends to submit this work for it. The author declares no other competing interests.

Data availability

Code, raw outputs and logs for every number reported here are at <https://github.com/jaideepsaipadhi/zero-gradient-prize>, and are archived with the extended version on Zenodo, [doi:10.5281/zenodo.XXXXXXXX](https://doi.org/10.5281/zenodo.XXXXXXXX).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) in order to develop and check proofs, write and run the numerical code, search and verify the literature, and draft and edit the text. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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